Program-2 : Implementation of Logistic Regression

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# Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix

# Create a sample dataset
data = {
    'Feature1': [2, 3, 5, 7, 9, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28],
    'Feature2': [1, 4, 6, 8, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 30],
    'Target':   [0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1]  # Binary classification (0 or 1)
}

# Convert dataset to DataFrame
df = pd.DataFrame(data)

# Split features and target
X = df[['Feature1', 'Feature2']]
y = df['Target']

# Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train Logistic Regression model
model = LogisticRegression()
model.fit(X_train, y_train)

# Make predictions
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```
y_pred = model.predict(X_test)

# Evaluate model
print(f"Accuracy: {accuracy_score(y_test, y_pred):.2f}")
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))

# Plot data visualization
plt.figure(figsize=(8, 6))
plt.scatter(df['Feature1'], df['Feature2'], c=df['Target'], cmap='bwr', edgecolors='k')
plt.xlabel('Feature1')
plt.ylabel('Feature2')
plt.title('Logistic Regression Data Visualization')
plt.colorbar(label="Target (0 = Blue, 1 = Red)")
plt.show()
```

OUTPUT-

```
Accuracy: 1.00
Confusion Matrix:
 [[1 0]
 [0 2]]
```